



EXPERIMENT 12

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EXPERIMENT: Advanced SQL Queries

AIM: To study and implement PostgreSQL triggers and trigger functions for automatically maintaining an audit log of employee INSERT and DELETE operations along with the current date and time.

Question 1: Design and implement a PostgreSQL trigger on the student table that automatically displays the details of the row whenever a record is inserted or deleted. The output should display the Student ID, Name, Age, and Class using the appropriate NEW and OLD references.

Solution: CREATE TABLE student (

id SERIAL PRIMARY KEY,

name VARCHAR(100),

age INT,

class VARCHAR(20)

);

-- STEP 2: Create Trigger Function

CREATE OR REPLACE FUNCTION fn_student_audit()

RETURNS TRIGGER

LANGUAGE plpgsql

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AS

\$\$

BEGIN

IF TG_OP = 'INSERT' THEN

RAISE NOTICE

'Inserted Row -> ID: %, Name: %, Age: %, Class: %',

NEW.id,

NEW.name,

NEW.age,

NEW.class;

RETURN NEW;

ELSIF TG_OP = 'DELETE' THEN

RAISE NOTICE

'Deleted Row -> ID: %, Name: %, Age: %, Class: %',

OLD.id,

OLD.name,

OLD.age,

OLD.class;

RETURN OLD;

END IF;

RETURN NULL;

END;

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-- STEP 3: Create Trigger

```
CREATE TRIGGER trg_student_audit  
AFTER INSERT OR DELETE  
ON student  
FOR EACH ROW  
EXECUTE FUNCTION fn_student_audit();
```

```
INSERT INTO student (name, age, class)  
VALUES ('Aman', 19, '11th');  
INSERT INTO student (name, age, class)  
VALUES ('Rohit', 20, '12th');  
SELECT * FROM student;
```

```
DELETE FROM student  
WHERE name = 'Aman';
```

```
SELECT * FROM student;
```

OUTPUT:

Data Output Messages Notifications				
	id [PK] integer	name character varying (100)	age integer	class character varying (20)
1	1	Aman	19	11th
2	2	Rohit	20	12th

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Data Output Messages Notifications				
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	id [PK] integer	name character varying (100)	age integer	class character varying (20)
1	2	Rohit	20	12th

Question 2: A company maintains employee information in tbl_employee and an audit log in tbl_employee_audit. Design and implement a PostgreSQL trigger such that:

- Whenever a new employee is inserted, an audit record is created with the message: "Employee name <emp_name> has been added at <current_time>"
- Whenever an employee is deleted, an audit record is created with the message: "Employee name <emp_name> has been deleted at <current_time>"
- The solution must use a PostgreSQL trigger function and trigger.
- Test the trigger using INSERT and DELETE operations and display the audit records.

Solution: CREATE TABLE tbl_employee (

emp_id SERIAL PRIMARY KEY,

emp_name VARCHAR(100) NOT

NULL,

emp_salary NUMERIC

);

CREATE TABLE tbl_employee_audit (

sno SERIAL PRIMARY KEY,

message TEXT

);

CREATE OR REPLACE FUNCTION audit_employee_changes()

RETURNS TRIGGER

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```
LANGUAGE plpgsql
AS
$$
BEGIN
    IF TG_OP = 'INSERT' THEN
        INSERT INTO tbl_employee_audit(message)
        VALUES (
            'Employee name ' || NEW.emp_name ||
            ' has been added at ' || NOW()
        );
        RETURN NEW;
    ELSIF TG_OP = 'DELETE' THEN
        INSERT INTO tbl_employee_audit(message)
        VALUES (
            'Employee name ' || OLD.emp_name ||
            ' has been deleted at ' || NOW()
        );
        RETURN OLD;
    END IF;
    RETURN NULL;
END;
$$;

CREATE TRIGGER trg_employee_audit
```

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AFTER INSERT OR DELETE

ON tbl_employee

FOR EACH ROW

EXECUTE FUNCTION audit_employee_changes();

INSERT INTO tbl_employee (emp_name, emp_salary)

VALUES ('Aman', 50000);

INSERT INTO tbl_employee (emp_name, emp_salary)

VALUES ('Rohit', 55000);

SELECT * FROM tbl_employee;

DELETE FROM tbl_employee

WHERE emp_name = 'Aman';

SELECT * FROM tbl_employee;

SELECT * FROM tbl_employee_audit;

OUTPUT:

Data Output Messages Notifications			
			
	emp_id [PK] integer	emp_name character varying (100)	emp_salary numeric
1	1	Aman	50000
2	2	Rohit	55000

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Data Output	Messages	Notifications
emp_id [PK] integer	emp_name character varying (100)	emp_salary numeric
1	2 Rohit	55000

Data Output	Messages	Notifications
sno [PK] integer	message text	
1	1	Employee name Aman has been added at 2026-09-26 00:24:25.750023+05:...
2	2	Employee name Rohit has been added at 2026-09-26 00:24:30.770569+05:...
3	3	Employee name Aman has been deleted at 2026-09-26 00:24:58.767164+0...

LEARNING OUTCOMES:

1. Understand the concept and purpose of PostgreSQL triggers and trigger functions.
2. Create PL/pgSQL trigger functions to automatically perform actions after database operations.
3. Use NEW, OLD, and TG_OP to handle INSERT and DELETE operations.
4. Implement an automated audit mechanism to record employee changes with timestamps.
5. Test and verify trigger execution by performing insert/delete operations and retrieving audit records.